Group 12

DRL Tracker
Software Development Plan (Small Project)
Version <1.0>

Revision History

Date	Version	Description	Author
<dd/mmm/yy>	<x.x>	<details>	<name>

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Software Development Plan (Small Project)

1. Introduction

1.1 Purpose

The purpose of this Software Development Plan is to define the development approach, project schedule, task assignments, and monitoring activities for DRL Tracker. It helps the Team Leader coordinate the project and enables team members to understand their responsibilities, deadlines, and expected deliverables.

1.2 Scope

This plan describes the overall development process for DRL Tracker based on the product needs, features, and non-functional requirements defined in the Vision Document. It covers project organization, scheduling, task allocation, progress monitoring, risk management, and deliverables.

1.3 Overview

This document describes the project overview, including its purpose, scope, objectives, assumptions, constraints, and deliverables. It also presents the project organization, development schedule, resource allocation, and processes for monitoring progress, managing risks, and controlling changes.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

DRL Tracker is a centralized system for managing university extracurricular activities and student conduct points. Based on the Vision Document, the project aims to support activity registration, evidence submission and verification, predicted score tracking, and the review and approval of official scores.

The main objectives are to:

- Provide students with activity discovery, registration, evidence management, and transparent score tracking.
- Enable Organizations to manage activities, participants, and evidence verification.
- Support faculty level Assessment Officers in reviewing records and proposing scores, and university level Assessment Officers in approving final Official Scores.
- Allow Admins to manage accounts, permissions, and training point regulations.
- Deliver a working system with supporting documentation and testing results.

2.2 Assumptions and Constraints

Assumptions

- The requirements and high level features defined in the Vision Document serve as the basis for development planning.
- Applicable university training point regulations and assessment procedures will be available for system configuration.
- Team members will collaborate and complete assigned tasks according to the agreed schedule.

Constraints

- The project must comply with course requirements and submission deadlines.
- Development resources are limited to three team members.
- The project is planned from 5 October to 13 December 2026, using five two week sprints.
- The implementation scope may be adjusted according to project progress and technical limitations.

2.3 Project Deliverables

The project is expected to produce:

- **Planning documents:** Project Proposal, Software Development Plan, and project tracking reports.
- **Requirements and design artifacts:** Vision Document, use cases, UI prototypes, and system design specifications.
- **Software product:** A working DRL Tracker implementation covering the agreed core features.
- **Testing artifacts:** Test cases, test results, and defect records.
- **Final submission materials:** Integrated software, updated documentation, and demonstration materials.

Deliverables will be developed throughout the project schedule, with final submission targeted for 13 December 2026.

3. Project Organization

3.1 Organizational Structure

The DRL Tracker project is built by a 3 member team: Minh, Tú, and Vĩ. Minh is the Team Leader and guides the team throughout the project.

The team works together, with each member helping in project planning, requirements, system design, coding, testing, and writing documents based on their assigned tasks. Team members talk regularly to discuss progress, solve problems, and make sure the work meets the agreed requirements and schedule.

The project uses parts of both RUP and Scrum. Development is split into three RUP phases: Inception, Elaboration, and Construction, organized into 5 sprints of 2 weeks each. The team uses the project schedule to assign tasks, work together, and track progress across these phases and sprints.

3.2 Roles and Responsibilities

Person	Role	Responsibilities
Minh	Team Leader, Requirements Analyst, Developer	Coordinates team activities, assigns tasks, monitors project progress, and consolidates weekly reports and project documentation. Leads requirements analysis and defines the functional scope of DRL Tracker. Participates in software development, integration, and debugging.
Tú	System Designer, Developer	Leads system design and technical architecture, including application structure, component interactions, and API design. Participates in requirements clarification, database integration, software development, integration, and debugging.
Vĩ	Database and Testing Coordinator, Developer	Leads database design and data management, prepares test cases, and coordinates software testing and defect tracking. Participates in requirements clarification, software development, integration, debugging, and technical documentation.

All team members are responsible for implementing their assigned features, reviewing relevant code, integrating components, and resolving technical issues collaboratively. Task assignments and deadlines are managed through the project schedule and reviewed regularly to ensure that project milestones are met.

4. Management Process

4.1 Project Estimates

The project schedule is estimated based on the course timeline, assignment deadlines, project scope, and the availability of the three team members. The project follows three RUP phases, organized into 5 sprints of 2 weeks each, covering project planning, requirements analysis, system design, implementation, and testing.

Task estimates and deadlines are managed through Jira. Estimates and task assignments may be reviewed when requirements change, technical dependencies arise, or actual progress differs from the planned schedule.

4.2 Project Plan

4.2.1 Phase Plan

The DRL Tracker project follows the RUP and Scrum. The project is organized into three RUP phases: Inception, Elaboration, and Construction, organized into 5 sprints of 2 weeks each.

The overall phase and sprint schedule is presented below.

RUP Phase	Sprint	Period	Assignment	Main objectives
Inception	1	5/10–18/10	PA0, PA1	Project proposal and initial project planning.
Elaboration	2	19/10–1/11	PA2	Requirements analysis and initial system design.
Elaboration	3	2/11–15/11	PA3	Detailed design, test planning, and working software.
Construction	4	16/11–29/11	PA4	Software implementation and testing.
Construction	5	30/11–13/12	PA5	Further implementation, integration, testing, and refinement.

The project milestones are associated with the completion of each sprint and its corresponding assignment. The team will review the requirements, deliverables, and task status at the end of each sprint to determine whether the planned objectives have been achieved.

The schedule may be revised when requirements change or implementation issues affect the planned activities.

4.2.2 Iteration Objectives

The team plans to produce three main software builds during the Elaboration and Construction phases. Each build will be used to evaluate the current implementation, identify defects, and guide subsequent development.

- **Build 1 - Initial Functional Build:** Planned for the end of Sprint 3 (15/11). It will provide an initial implementation of the core system structure and selected features for preliminary testing.
- **Build 2 - Integrated Build:** Planned for the end of Sprint 4 (29/11). It will incorporate additional implemented features and support integration and functional testing.
- **Build 3 - Final Build:** Planned for the end of Sprint 5 (13/12). It will include the completed in-scope features, integration fixes, and final testing in preparation for the project demonstration or submission.

The build schedule may be adjusted according to implementation progress and the assignment requirements.

4.2.3 Releases

The project schedule is managed using Jira to organize tasks, assign responsibilities, set deadlines, and monitor progress across the five sprints. Each sprint contains tasks related to its assigned project objectives and course assignments.

The screenshots below show the current task schedule for Sprint 1 and the preliminary plans for subsequent sprints. Tasks from Sprint 2 onward are tentative and have not yet been fully assigned to individual team members. They will be further refined and assigned as the project progresses. The schedule may be updated based on changes in requirements, team discussions, and actual progress.

Sprint 1: Inception (5/10 - 18/10)

Sprint 1 focuses on project initiation and the completion of PA0 and PA1. Activities include project proposal preparation, user interviews, UI prototyping, project planning, and weekly reporting.

Summary	Sprint	Assignee	Created	Updated	Due date ↑	Description	Status
Phỏng vấn user 2	Sprint 1	TT Tú Dươn...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 10, 2026	None	Done
Phỏng vấn user 1	Sprint 1	VP Vĩ Phạm...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 10, 2026	Phỏng vấn các user tiềm năng	Done
Tool setup	Sprint 1	MT Minh Ng...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 11, 2026	None	Done
PA0	Sprint 1	MT Minh Ng...	Oct 11, 2026,...	Oct 18, 2026, ...	Oct 11, 2026	Kiểm tra và nộp PA0	Done
Seminar 1	Sprint 1	MT Minh Ng...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 15, 2026	Chuẩn bị script	Done
Project Execution and Tracking	Sprint 1	MT Minh Ng...	Oct 18, 2026...	Oct 18, 2026, ...	Oct 18, 2026	None	Done
PA1	Sprint 1	MT Minh Ng...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	Kiểm tra lần cuối và nộp PA1.	Done

Summary	Sprint	Assignee	Created	Updated	Due date	Description	Status
UI prototyping	Sprint 1	Vĩ Phạm...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	None	Done
rup_sdpln	Sprint 1	Minh Ng...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	Thực hiện mục 3,4	Done
rup_sdpln	Sprint 1	Tú Dươn...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	Thực hiện mục 1,2	Done
rup_vision	Sprint 1	Vĩ Phạm...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	Thực hiện mục 4,5,6	Done
rup_vision	Sprint 1	Tú Dươn...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	Thực hiện mục 1,2,3	Done
Weekly report 2	Sprint 1	Minh Ng...	Oct 14, 2026...	Oct 18, 2026, ...	Oct 18, 2026	None	Done
Weekly report 1	Sprint 1	Minh Ng...	Oct 11, 2026,...	Oct 18, 2026, ...	Oct 18, 2026	None	Done

Sprint 2 – Elaboration (19/10 - 01/11)

Sprint 2 focuses on requirements analysis and initial system design, supporting the completion of PA2.

Summary	Sprint	Assignee	Created	Updated	Du...	Description	Status
Weekly report	Sprint 2	Minh Ng...	Oct 18, 2026, 3...	Oct 18, 2026, ...	Nov 1, 2026	None	To Do
PA2	Sprint 2	Minh Ng...	Oct 18, 2026, 1:...	Oct 18, 2026, ...	Nov 1, 2026	Kiểm tra và hoàn thành PA2	To Do
Create initial UI designs and navigation flows	Sprint 2	Unassig...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Oct 31, 2026	Thiết kế giao diện ban đầu	To Do
Design initial database model	Sprint 2	Vĩ Phạm...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Oct 28, 2026	Thiết kế mô hình cơ sở dữ liệu ban đầu	To Do
Design initial system architecture	Sprint 2	Tú Dươn...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Oct 28, 2026	Sơ đồ kiến trúc và các thành phần chính	To Do
Functional requirements and system scope	Sprint 2	Minh Ng...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Oct 21, 2026	Danh sách chức năng và phạm vi hệ thống.	To Do

Sprint 3 – Elaboration (02/11 - 15/11)

Sprint 3 focuses on detailed system design, test planning, and the preparation of working software, supporting the completion of PA3.

Summary	Sprint	Assignee	Created	Updated	Du...	Description	Status
Finalize detailed use cases and business rules	Sprint 3	Minh Ng...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Nov 4, 2026	Đặc tả chi tiết các luồng nghiệp vụ chính	To Do
Finalize database schema and relationships	Sprint 3	Vĩ Phạm...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Nov 7, 2026	Schema dữ liệu đủ để bắt đầu triển khai	To Do
Design detailed architecture and API contracts	Sprint 3	Tú Dươn...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Nov 7, 2026	Thiết kế API, các thành phần và cách giao...	To Do
Prepare test plan and test cases	Sprint 3	Unassig...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Nov 9, 2026	Kế hoạch kiểm thử và các trường hợp kiể...	To Do
Implement initial application structure	Sprint 3	Unassig...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Nov 12, 2026	Khung ứng dụng, cấu hình và kết nối nền t...	To Do
Implement and test a core workflow prototype	Sprint 3	Unassig...	Oct 18, 2026, 4...	Oct 18, 2026, ...	Nov 14, 2026	Ít nhất một luồng nghiệp vụ cơ bản có thể...	To Do
Weekly report	Sprint 3	Minh Ng...	Oct 18, 2026, 3...	Oct 18, 2026, ...	Nov 15, 2026	None	To Do
PA3	Sprint 3	Minh Ng...	Oct 18, 2026, 1:...	Oct 18, 2026, ...	Nov 15, 2026	Kiểm tra và hoàn thành PA3	To Do

Sprint 4 – Construction (16/11 - 29/11)

Sprint 4 focuses on software implementation and testing, supporting the completion of PA4.

Summary	Sprint	Assignee	Created	Updated	Du...	Description
Implement authentication and role-based access	Sprint 4	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 19, 2026	Đăng nhập và phân quyền cơ bản
Implement student activity browsing and registration	Sprint 4	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 22, 2026	Student có thể xem và đăng ký hoạt động
Implement evidence submission and tracking	Sprint 4	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 23, 2026	Student có thể nộp và theo dõi minh chứng
Implement organization activity management and evidence review	Sprint 4	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 24, 2026	Organization quản lý hoạt động và xử lý minh chứng
Implement database integration for core features	Sprint 4	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 25, 2026	Các chức năng chính kết nối và lưu dữ liệu
Integrate core modules and perform functional testing	Sprint 4	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 27, 2026	Các luồng chính hoạt động, có kết quả kiểm thử
Weekly report	Sprint 4	MT Minh Ng...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 29, 2026	None
PA4	Sprint 4	MT Minh Ng...	Oct 18, 2026...	Oct 18, 2026, ...	Nov 29, 2026	Kiểm tra và hoàn thành PA4

Sprint 5 – Construction (30/11 - 13/12)

Sprint 5 focuses on further software implementation, integration, testing, and refinement, supporting the completion of PA5.

Summary	Sprint	Assignee	Created	Updated	Due date	Description
Implement assessment workflow for faculty and university officers	Sprint 5	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 3, 2026	Luồng rà soát, đề xuất và phê duyệt kết quả đánh giá
Implement scoring rules and official score calculation	Sprint 5	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 5, 2026	Tính điểm theo quy chế và lưu kết quả phù hợp
Implement admin functions and remaining in-scope features	Sprint 5	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 6, 2026	Hoàn thiện các chức năng quản trị được yêu cầu
Integrate assessment workflow with evidence and activity data	Sprint 5	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 8, 2026	Các luồng nghiệp vụ kết nối với nhau
Perform end-to-end and regression testing	Sprint 5	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 10, 2026	Kiểm thử luồng xuyên suốt và kiểm tra lỗi hồi quy
Fix defects and verify acceptance criteria	Sprint 5	Unassig...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 11, 2026	Xử lý lỗi và xác nhận tiêu chí nghiệm thu
Weekly report	Sprint 5	MT Minh Ng...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 13, 2026	None
PA5	Sprint 5	MT Minh Ng...	Oct 18, 2026...	Oct 18, 2026, ...	Dec 13, 2026	Kiểm tra và hoàn thành PA5

4.2.4 Project Schedule

The overall project schedule follows the course timeline and assignment deadlines. The project consists of five two week sprints across the Inception, Elaboration, and Construction phases, from 5 October to 13 December.

Jira is used to maintain the detailed project schedule, including task descriptions, assignees, deadlines, dependencies, and progress status. The schedule is reviewed regularly by the Team Leader and updated when changes are agreed upon by the team.

The overall schedule and sprint-level tasks are presented in Section 4.2.3.

4.2.5 Project Resourcing

The project is carried out by three team members: Minh, Tú, and Vĩ. All members participate in software development, integration, and debugging.

Minh is responsible for team coordination, task assignment, progress monitoring, report consolidation, and requirements analysis. Tú leads system design and technical architecture, while Vĩ leads database design and testing coordination. These responsibilities define the primary areas of ownership, while implementation tasks are shared according to the project schedule.

The team uses Jira to allocate tasks and monitor workload and deadlines.Team members will share technical knowledge and coordinate their work to address skill gaps and implementation dependencies.

4.3 Project Monitoring and Control

4.3.1 Requirements Management

System requirements are documented in the DRL Tracker system overview and related project documents. Jira is used to track implementation tasks, assigned team members, deadlines, and progress.

New or modified requirements are discussed during team meetings to clarify their scope, expected behavior, and impact on existing features and the project schedule. Following Seminar 1, the team identified the need to refine certain system details and functional scopes. The team therefore discusses and agrees on these changes before updating the relevant documentation and Jira tasks.

Minh coordinates requirements analysis and maintains consistency between the agreed scope and project plan. Tú and Vĩ contribute to evaluating technical implications, data requirements, and testing considerations. All members are responsible for implementing the approved requirements assigned to them.

The team reviews requirements throughout each sprint to ensure that the planned and implemented functionality remains consistent with the agreed project scope.

4.3.2 Reporting and Measurement

4.3.3 Risk Management

The team identifies and assesses project risks during the Inception Phase. The risk register is reviewed during team meetings and updated when new risks emerge or existing risks change.

Risk Ranking	Risk Description and Impact	Mitigation Strategy and/or Contingency Plan
High	Changing or unclear requirements may cause rework and delay project milestones.	Clarify requirements during team meetings, document agreed decisions, and update Jira tasks and deadlines.
High	Delays in assigned tasks may affect dependent tasks and sprint deliverables.	Monitor progress regularly, identify blocked tasks early, and redistribute work when necessary.
Medium	Integration issues may occur when combining software components developed by different members.	Agree on interfaces and data structures early, integrate incrementally, and test components throughout development.
Medium	Testing may reveal defects or incomplete functionality near assignment deadlines.	Prepare test cases in advance, test incrementally, prioritize critical defects, and reserve time for fixes.

Medium	Limited team capacity may make it difficult to complete all planned features within the available schedule.	Prioritize essential features, review workload at each sprint, and agree on scope adjustments when needed.

4.3.4 Configuration Management

The team uses Facebook Group as the primary communication channel, Google Docs for collaborative document editing, GitHub for storing completed documents and managing source code versions, and Jira for tracking tasks and issues.

- **Change and Issue Management:** Problems and proposed changes are discussed within the Facebook Group and tracked through Jira when necessary. The team reviews their impact, agrees on actions, and assigns tasks to responsible members. Changes are implemented and verified before being incorporated into the project.
- **Naming and Versioning:** Documents use descriptive filenames and revision dates or version numbers when appropriate. Source code is managed through GitHub commits, while test cases and results are stored with their related project artifacts.
- **Storage, Backup, and Recovery:** Working documents are maintained in Google Docs and uploaded to GitHub when completed. Important files should be retained until project assessment is complete, with additional backups made when necessary.
- **Media and Access:** Project artifacts are stored digitally in Google Docs, GitHub, and Jira. Access to shared resources is limited to team members, and documents are maintained in commonly accessible formats such as Google Docs, PDF, and source code files.